Sandro Martinelli Reia, PhD

Geospatial Data Analyst | Urban Physics | Python-first applied scientist

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Summary

Applied researcher and data analyst with 10+ years of experience turning complex, heterogeneous datasets into reproducible analyses, models, and decision-ready outputs. Strong Python-based statistical computing, geospatial analytics, and scalable workflow development across mobility, urban systems, and disaster recovery. Published 30+ peer-reviewed papers (including *Nature Communications* and *Nature Cities*) and experienced collaborating across interdisciplinary teams.

Core Skills

Programming & Stats	Expert Python (Pandas, NumPy, SciPy, Statsmodels). Advanced multivariate analysis, hypothesis testing, and regression modeling. Architect of end-to-end reproducible research workflows.
Geospatial & GIS	Specialized in GeoPandas, Shapely, and PyProj. Expert in spatial joins, network analysis, and the integration of heterogeneous spatial, tabular, and temporal datasets.
Machine Learning	Applied supervised learning for classification and regression. Proficient in feature engineering, model evaluation/validation, and managing large-scale labeled datasets.
Data Sources	Deep experience with OpenStreetMap, building footprints, mobility/trajectory data, POI foot-traffic signals, and longitudinal demographic datasets.
Leadership & Mgmt	Supervised PhD students and led day-to-day laboratory activities. Expert in project lifecycle management, interdisciplinary team coordination, and technical mentorship.
Communication	Technical scientific writing with 30+ publications. Designer of stakeholder-facing visualizations and maps. Skilled at translating complex data insights for non-technical audiences.

Professional experience

Postdoctoral Researcher, Geography & Geoinformation Science — 2023 – Present, Fairfax, VA

George Mason University

- Build end-to-end analysis pipelines for large-scale urban mobility simulation and validation, integrating heterogeneous inputs (e.g., buildings/land use, networks, and behavioral models) into consistent data products.
- Develop scalable Python workflows for data ingestion, quality control, and downstream analytics; emphasize reproducibility (documented assumptions, well-defined outputs, and consistency checks).
- Collaborate with interdisciplinary teams to translate research questions into analyzable datasets, interpretable visualizations, and written technical reports.

Postdoctoral Researcher, Civil Engineering — 2021 – 2023, West Lafayette, IN

Purdue University

- Conducted statistical analyses and model-based studies of human mobility and resilience/recovery dynamics in interdependent social and physical systems.
- Integrated multi-source datasets (mobility, infrastructure, and contextual covariates) into coherent analytical datasets; applied multivariate methods to quantify drivers and outcomes.
- Delivered peer-reviewed research outputs and communicated results to collaborators across domains.

Postdoctoral Researcher (Fellow), Institute of Physics — 2016 – 2021, São Paulo, Brazil

University of São Paulo (USP) / Indiana University (IU)

- Applied NLP and machine learning to analyze the diffusion of innovations across large-scale text archives and citation databases.
- Designed relational databases to manage longitudinal study data, ensuring data integrity and efficient querying.
- Secured $120k+ in competitive research grants.

Selected projects

IARPA HAYSTAC — large-scale urban mobility simulation (main developer)

End-to-end simulation and analysis pipeline for urban mobility patterns-of-life research (IARPA HAYSTAC)

- Lead developer for a scalable urban mobility simulator supporting synthetic population generation, activity scheduling, and trip/mode decisions with geospatially grounded inputs.
- Built reproducible data pipelines integrating heterogeneous sources (e.g., buildings/land use, networks, and mobility resources) with rigorous validation checks and well-documented outputs.

Disaster Infrastructure Recovery Simulator (DIRSim)

Agent-based analysis of post-disaster recovery using multilayer socio-physical networks and real-world mobility/POI-derived signals

- Simulated recovery dynamics and evaluated policy scenarios; highlighted heterogeneous recovery pathways relevant to climate adaptation and community resilience.

OpenStreetMap-derived U.S. building classification dataset (*Scientific Data*)

Large-scale geospatial dataset creation, validation, and release for broad research and practitioner use

- Contributed to dataset engineering and analytical framing; supported downstream uses in urban development and land-use applications.

Education

PhD in Sciences (Statistical/Computational Physics) — 2015, Brazil

University of São Paulo (USP)
Thesis: Activity, phase transition and media effect in a sociocultural model

MSc in Sciences (Physics) — 2011, Brazil

University of São Paulo (USP)
Dissertation: Phase transitions and nucleation processes in cellular automata rule space

BSc in Physics — 2008, Brazil

University of São Paulo (USP)

Selected publications (full list on Google Scholar)

- **Nature Cities** — *Domestic migration and city rank dynamics* (Reia et al., 2025)
- **Nature Communications** — *Spatial structure of city population growth* (Reia et al., 2022)
- **Scientific Data** — *OpenStreetMap derived building classification dataset for the United States* (Arruda et al., 2024)
- **Sustainable Cities and Society** — *Agent-based model of post-disaster recovery in multilayer socio-physical networks* (Xue et al., 2024)
- **Computers, Environment and Urban Systems** — *Function and form of US cities* (Reia et al., 2025)

Certificates

- Statistics for Data Science: Theory and Practice with Python (ICMC/USP, 2021)
- Machine Learning (Stanford Online via Coursera, 2021)
- Programming with Python for Data Science (Microsoft/edX, 2017)

Languages

- Portuguese (native)
- English (Cambridge CPE)

Academic service

Academic Editor, PLOS ONE (Urban Studies, Complexity, Networks)
Reviewer for journals including CEUS, IJGIS, Scientific Reports, and others.